

H8 真实材料公式兼容性

1. fraction-radical

$$\frac{\sqrt{2}+1}{2}$$

2. nested-fraction

$$\frac{3}{\tan A} + \frac{2}{\tan B} = 1$$

3. triangle-angle

$$\triangle ABC, \angle ABC = 120^\circ$$

4. parallel-perpendicular

$$AB \parallel CD, AD \perp AB$$

5. vector-short

$$\vec{a} + \vec{b}$$

6. vector-segment

$$\overrightarrow{AB} = \vec{a} + m\vec{b}$$

7. vector-dot-product

$$\overrightarrow{CP} \cdot \overrightarrow{DP}$$

8. overline-segment

$$\overline{DM} \cdot \overline{DN} = 18 - 6\sqrt{2}$$

9. set-builder

$$A = \{x \mid x = m + \sqrt{3}n, m \in \mathbb{Z}, n \in \mathbb{Z}\}$$

10. set-operations

$$A \cup B, A \cap B, A \subseteq B$$

11. interval

$$x \in \left(\frac{\pi}{4}, \pi\right)$$

12. absolute-value

$$y = |f(x-1)| + 1$$

13. piecewise

$$f(x) = \begin{cases} x^2 + 2x - 3, & x \leq 0 \\ -2 + \ln x, & x > 0 \end{cases}$$

14. matrix

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

15. array

$$\begin{aligned}x + y &= 1 \\x - y &= 3\end{aligned}$$

16. summation

$$\sum_{i=1}^n (x_i - \bar{x})^2 = 10$$

17. correlation

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

18. chi-square

$$\chi^2 = \frac{n(ad - bc)^2}{(a + b)(c + d)(a + c)(b + d)}$$

19. binomial

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

20. probability

$$P(A \mid B) = \frac{P(AB)}{P(B)}$$

21. limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

22. derivative

$$f'(x) = \frac{\mathrm{d}}{\mathrm{d}x} f(x)$$

23. integral

$$\int_a^b f(x) \mathrm{d}x$$

24. logarithm

$$\log_a(MN) = \log_a M + \log_a N$$

25. trigonometric

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

26. ellipse

$$\frac{x^2}{m} + \frac{y^2}{n} = 1 \quad (m > n > 0)$$

27. coordinates

$$A(0, 0), B(3, 0), C(2, \sqrt{3}), D(0, \sqrt{3})$$

28. complex

$$z = a + bi, \quad i^2 = -1$$

29. sequence

$$S_n = \frac{n(a_1 + a_n)}{2}$$

30. recurrence

$$a_{n+1} = 2a_n + 1$$

31. subscript-superscript

$$P_1(1, 1), P_2(-1, 1), P_3(-1, -1), P_4(1, -1)$$

32. max-min

$$\max_{0 \leq x \leq 3} (x^2 - 2x + 3)$$